MIL-B-6038A
28 MARCH 1961
Superseding
MIL-B-6038
15 March 1950

MILITARY SPECIFICATION

BEARINGS, BALL, BELLCRANK, ANTIFRICTION, AIRFRAME

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 This specification covers antifriction bellcrank ball bearings intended primarily for airframe use.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

QQ-P-416—Plating, Cadmium (Electrodeposited)
QQ-S-624—Steel Bars, Alloy, Cold Finished and Hot Rolled (General Purpose)
QQ-S-633—Steel Bars, Carbon, Cold Finished and Hot Rolled (General Purpose)

MILITARY

MIL-P-197—Preservation, Packaging and Packing of Antifriction Bearings, Associated Parts, and Sub-Assemblies
MIL-B-1083—Balls, Bearing, Alloy Steel
MIL-G-3278—Grease; Aircraft and Instrument (for Low and High Temperatures)
MIL-E-5272—Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-S-7420—Steel Bars, Alloy, Chromium, High Carbon E52100 (Aircraft Quality)

STANDARDS

FEDERAL

Federal Standard No. 66—Steel: Chemical Composition and Hardenability

MILITARY

MIL-STD-10—Surface Roughness, Waviness, and Lay
MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129—Marking for Shipment and Storage
MIL-STD-130—Identification Marking of U.S. Military Property
MS20218—Bearings, Ball, Bellcrank, Antifriction, Airframe

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 **Other publications.** The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

CONSOLIDATED CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules

(Application for copies should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station, Chicago 6, Illinois.)

3. REQUIREMENTS

3.1 **Preproduction.** This specification makes provisions for preproduction testing (see 6.2).

3.2 **Materials.** The bearing rings and balls shall be made of steel conforming to Specification MIL-S-7420, QQ-S-624, QQ-S-633, and Federal Standard No. 66, steel numbers 50100, 51100, and 52100. Balls shall conform to Specification MIL-B-1083.

3.3 **Design and construction.** The bearings shall be designed and constructed to conform

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to the requirements specified herein and on Standard MS20218.

3.3.1 *Dimensions and tolerances.* The dimensions and tolerances shall be as specified on Standard MS20218.

3.3.2 *Hardness.* The raceways shall be uniformly through-hardened 58–62 Rockwell "C" scale. The ball hardness shall be in accordance with Specification MIL–B–1083. The hardness of the riveting flanges of the bearings shall be not greater than Rockwell C–25 (see 4.5.2).

3.3.3 *Surface roughness.* The raceways shall be ground or lapped to a surface finish of 4 microinches maximum in accordance with Standard MIL–STD–10. The bore and all external surfaces shall have a surface finish of 32 microinches maximum. The balls shall conform to grade 2 of Specification MIL–B–1083.

3.3.4 *Plating.* All bearing external surfaces and the bore shall be plated. The plating shall be in accordance with class 2, type I of Specification QQ–P–416. The plating thickness shall not exceed 0.0005 inch maximum (see 4.5.4).

3.3.5 *Seals.* Closures shall be provided to prevent the entrance of contaminants and to retain the lubricant throughout the temperature range specified in 3.3.5.1.

3.3.5.1 *Type.* The closures shall be of the positive contact type, consisting of external metal shields and positive seal washers that shall operate satisfactorily when tested as specified in 4.5.5. The closures shall be readily removable for inspection and relubrication and shall be of such construction that they may be replaced without undue difficulty. The closures shall withstand temperatures of −67° to 250° F without warpage or embrittlement that will prevent retention of the lubricant. The closures shall be compatible with the lubricant specified herein.

3.3.6 *Retainers.* The retainers shall be of a suitable material consistent with the requirements of this specification. They shall be so constructed as to not limit the proper functioning of the bearings for which they are designed.

3.4 Internal clearance.

3.4.1 *Radial play.* The radial play, measured before and after the radial play and axial static capacity tests (4.5.6 and 4.5.10, respectively), shall not exceed the values specified on Standard MS20218.

3.4.2 *Axial play.* The axial play, measured before and after the radial play and axial static capacity tests (4.5.6 and 4.5.10, respectively), shall not exceed the values specified on Standard MS20218 (see 4.5.7).

3.5 Radial strength.

3.5.1 *Radial static capacity.* The radial static capacity of the bearing shall be not less than the value specified on Standard MS20218.

3.5.2 *Radial fracture load.* The radial fracture load of the bearing shall be not less than one and one-half times the radial static capacity value specified on Standard MS20218.

3.6 Axial strength.

3.6.1 *Axial static capacity.* The axial static capacity of the bearing shall be not less than the value specified on Standard MS20218.

3.6.2 *Axial fracture load.* The axial fracture load of the bearing shall be not less than one and one-half times the axial static capacity value specified on Standard MS20218.

3.7 Wobble. The total wobble, measured before and after the radial play and axial static capacity tests (4.5.6 and 4.5.10, respectively) shall be not greater than the value specified on Standard MS20218.

3.8 Radial endurance. The load applied to the bearing in the radial endurance test (4.5.13) shall be as specified on Standard MS20218.

3.9 Lubrication. Unless otherwise specified, the bearing shall be filled with grease conforming to Specification MIL–G–3278, completely filling the bearing voids. The bearing shall be thoroughly cleaned and dried before being lubricated.

3.10 Stabilization. The bearing components shall not vary more than 0.0001 inch dimensionally for operation from −67° to 250° F.

3.11 Interchangeability. All parts having the same MS number shall be both functionally and dimensionally interchangeable.

3.12 Identification of product. The bearings shall be permanently and legibly marked for identification in accordance with Standard MIL–STD–130. The identification data applied to the bearings shall be as follows:

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(a) Complete MS number
(b) Manufacturer's name or trademark
(c) U.S.

3.13 Workmanship. The bearings shall be free of tool marks, chatter waves, rust, grinding scratches, pits, or any other defects that may adversely affect their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Inspection responsibility. The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of bearings shall be classified as follows:

(a) Preproduction tests	(4.3)
(b) Acceptance tests	(4.4)

4.3 Preproduction tests.

4.3.1 *Sampling.* The test samples shall consist of 20 bearings of each MS number and shall be representative of the production equipment. The bearings shall be tested at a laboratory designated by the procuring activity or, when so stated in the contract, at the contractor's plant under the supervision of the procuring activity (see 6.2).

4.3.2 *Tests.* The preproduction tests shall consist of all the tests specified under 4.5.

4.4 Acceptance tests.

4.4.1 *Lot.* The lot definition, formation, and size shall be in accordance with Standard MIL–STD–105.

4.4.2 *Sampling.* The sample bearings shall be selected in accordance with the tables titled "Sample size code letters for ordinary inspection" and "Master table for normal and tightened inspection (single sampling)" at Inspection Level II of Standard MIL–STD–105. The lots shall be accepted or rejected in accordance with Acceptable Quality Level (AQL) of 1.0 percent defective on the basis of tests listed in 4.4.3.

4.4.3 *Tests.* The acceptance tests of the bearing shall consist of the following:

(a) Examination of product	(4.5.1)
(b) Hardness	(4.5.2)
(c) Surface roughness	(4.5.3)
(d) Radial play	(4.5.6)
(e) Axial play	(4.5.7)
(f) Wobble	(4.5.12)
(g) Lubrication	(4.5.14)

4.5 Test methods.

4.5.1 *Examination of product.* Each sample bearing shall be inspected for conformance to this specification and Standard MS20218 with respect to dimensions, identification, finish, lubrication, and workmanship.

4.5.2 *Hardness.* Two bearings of each MS number shall be disassembled and tested for hardness on the inner ring, the outer ring, and on three balls. The tests on the inner and outer rings shall be made on rolling raceways with a standard hardness tester, using the Rockwell "C" scale. The hardness shall be as specified in 3.3.2 and the balls shall be tested in accordance with Specification MIL–B–1083.

4.5.2.1 *Hardness of riveting flanges.* The hardness of the riveting flanges shall be tested, using the Rockwell "C" scale, and the hardness shall be as specified in 3.3.2.

4.5.3 *Surface roughness.* The surface roughness shall be measured in accordance with Standard MIL–STD–10. The test shall be made on the bearings disassembled for the tests specified in 4.5.2.

4.5.4 *Plating.* The plating shall be tested for conformance with 3.3.4 and Standard MS20218. Two bearings of each MS number shall be tested.

4.5.5 *Seals.* The seals shall be subjected to temperatures of −67° ±10° F and 275° ±10° F for 100 hours each. The seals shall then operate in a test chamber at 3 revolutions per minute (rpm) while tested in accordance with the high temperature tests, Procedure II and the low temperature tests, Procedure I of Specification MIL–E–5272. Seals that show warpage, embrittlement, nonretention of lubricant, incompatibility to Specification MIL–

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G-3278 grease, or ingress of dust into the bearing shall be considered seal failure. Three bearings of each MS number shall be tested (see 3.3.5).

4.5.6 *Radial play.* Five bearings of each MS number in the sample shall be tested for radial play. The inner ring shall be held rigidly and a radial load of 5½ pounds applied alternately in opposite directions to the outer ring. A dial indicator shall be used to indicate this movement. The raidal play is the total movement from application of a 5½-pound load on one side to the load application on the opposite side (see 3.4.1).

4.5.7 *Axial play.* Five bearings of each MS number in the sample shall be tested for axial play. One ring shall be held rigidly and an axial load of 5½ pounds applied alternately in opposite directions to the other ring. A dial indicator shall be used to indicate this movement. The radial play is the total movement from application of a 5½-pound load on one side to the load application on the opposite side (see 3.4.2).

4.5.8 *Radial static capacity.* The bearing shall be mounted in a rigid support as shown on figure 1 and rotated manually to determine operational smoothness before testing. The starting friction torque shall be measured. The appropriate static capacity load specified on Standard MS20218 shall be applied for 1 minute. The load shall be removed and the bearing inspected for evidence of failure. If the starting friction torque has increased 100 percent or more, the bearing shall be considered to have failed. Three bearings of each MS number shall be tested.

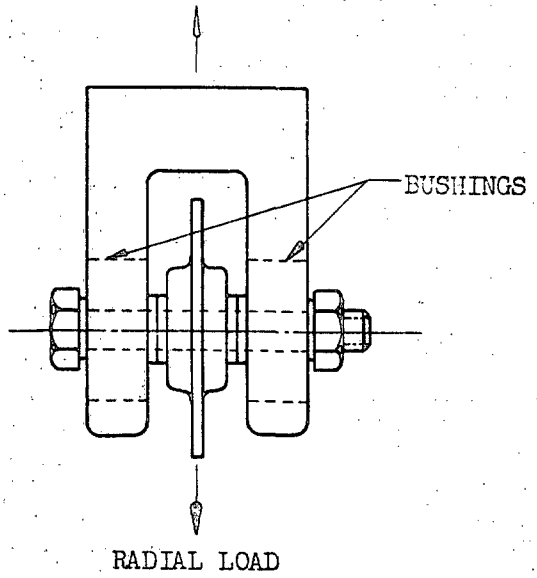


FIGURE 1. *Radial static capacity setup*

4.5.9 *Radial fracture load.* The radial fracture load of one and one-half times the pertinent radial static capacity value shown on Standard MS20218 shall be applied to the bearing for 1 minute. After removal of the load, the bearing shall be examined for failure. Bearings that cannot be turned by hand, or that have parts fractured, shall be considered to have failed. Three bearings of each MS number shall be tested.

4.5.10 *Axial static capacity.* The bearing shall be mounted in a rigid support as shown on figure 2 and rotated manually to determine operational smoothness before testing. The starting friction torque shall be measured. The appropriate static capacity load specified on Standard MS20218 shall be applied for 1 minute. The load shall be removed and the bearing inspected for evidence of failure. If the starting friction torque has increased 100 percent or more, the bearing shall be considered to have failed. Three bearings of each MS number shall be tested.

4.5.11 *Axial fracture load.* The axial fracture load of one-half times the pertinent axial static capacity value shown on Standard MS20218 shall be applied to the bearing for 1 minute. After removal of this load, the bear-

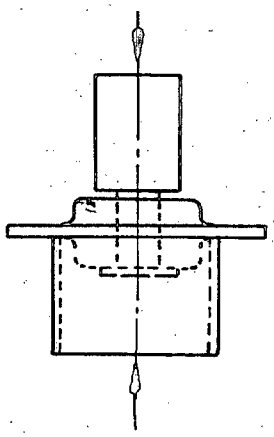


FIGURE 2. *Axial static test setup*

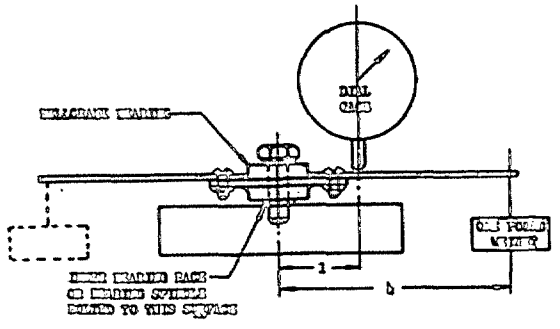
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ing shall be examined for failure. Bearings that cannot be turned by hand, or that have parts fractured, shall be considered to have failed. Three bearings of each MS number shall be tested.

4.5.12 *Wobble.* The bearing shall be tested for total wobble as indicated on figure 3. The wobble shall be measured at three points equally spaced about the bearing axis, and which lie in a plane that is parallel to the faces of the bearing flange. Ten bearings of each MS number shall be tested.

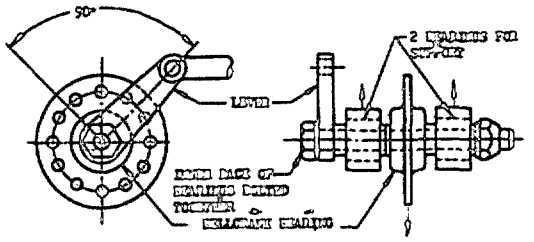
4.5.13 *Radial endurance.* The bearing shall be subjected to the radial endurance load specified on Standard MS20218 for 50,000 reversals of rotation at the rate of 40 reversals per minute. Each reversal shall consist of one-fourth of a revolution (90 degrees) of the bearing. The endurance test shall be made as indicated on figure 4, with the inner race of the bellcrank bearing bolted between the inner races of the two supporting bearings so that the inner races of the three bearings rotate as a unit. The oscillating motion shall be uniform throughout each reversal so that no impact load will be applied either at the beginning or end of a reversal. The specified radial endurance load shall be applied to the flange of the bearing in a manner that will not permit rotation of the outer race. The inner race of the bearing shall then be subjected to the required oscillating motion. Three bearings of each MS number shall be tested.

4.5.14 *Lubrication.* The bearing shall be inspected for quantity and cleanliness of the lubricant and tested in accordance with Specification MIL-G-3278, unless other lubricant is specified by the procuring activity.



DIMENSIONS IN INCHES. TOLERANCES ±0.10 INCH.

FIGURE 3. *Setup for wobble test*



OSCILLATING MOTION OF BEARING INNER RACES 90°.

FIGURE 4. *Endurance test setup*

4.5.15 *Stabilization.* The bearing shall be held at 275° ±10° F for 100 hours and then measured. The bearing shall then be cold soaked at −67° ±10° F for 100 hours and then measured. All measurements shall be made at room temperature. Bearings that show a dimensional change of more than 0.0001 inch per inch of diameter shall be considered to have failed. Three bearings of each MS number shall be tested.

4.5.16 *Cleaning, preservation, packaging, and packing.* Preparation for delivery shall be inspected for conformance to section 5.

5. PREPARATION FOR DELIVERY

5.1 Cleaning, preservation, and packaging (see 6.2).

5.1.1 *Level A.* Bearings shall be preserved and packaged in accordance with Specification MIL-P-197.

5.1.2 *Level C.* Cleaning, preservation, and packaging shall be in accordance with the contractor's commercial practice.

5.2 Packing (see 6.2).

5.2.1 *Level A.* Bearings, packaged level A or C as specified, shall be packed for overseas shipment in accordance with Specification MIL-P-197.

5.2.2 *Level B.* Bearings, packaged level A or B as specified, shall be packed for domestic shipment and storage in accordance with Specification MIL-P-197.

5.2.3 *Level C.* Bearings, packaged in accordance with level A or C, shall be packed in a manner to insure carrier acceptance and safe delivery at destination. The containers shall

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be in accordance with the Uniform Freight Classification Rules or regulations of other carriers as applicable to the mode of transportation.

5.3 Marking of shipments. Interior packages and exterior shipping containers shall be marked in accordance with Standard MIL-STD-129. The nomenclature shall include the following:

BEARINGS, BALL, BELLCRANK
ANTIFRICTION, AIRFRAME

6. NOTES

6.1 Intended use. The antifriction bearings covered by this specification are intended primarily for use in aircraft control systems. These bearings are not intended for applications involving high rotative speeds.

6.1.1 This specification contains procurement requirements for bearings shown on Standard MS20218.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Quantity and type required.
(c) Whether cleaning, preservation, and packaging should be level A or C, and packing level A, B, or C (see 5.1 and 5.2).
(d) Point of preproduction testing (see 3.1 and 4.3.1).

Notice: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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